INCH-POUND
MIL-R-83726/18H
11 January 2016
SUPERSEDING
MIL-R-83726/18G
15 June 2002

DETAIL SPECIFICATION SHEET

# RELAYS, HYBRID, TIME DELAY (ON RELEASE), TRACK MOUNTED, TYPE IIA, CLASS B, 10 AMPERES, 2PDT, HERMETICALLY SEALED, FIXED TIME, 0.1 TO 300 SECONDS

Inactive for new design after 23 June 1995.
No superseding document.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-83726.

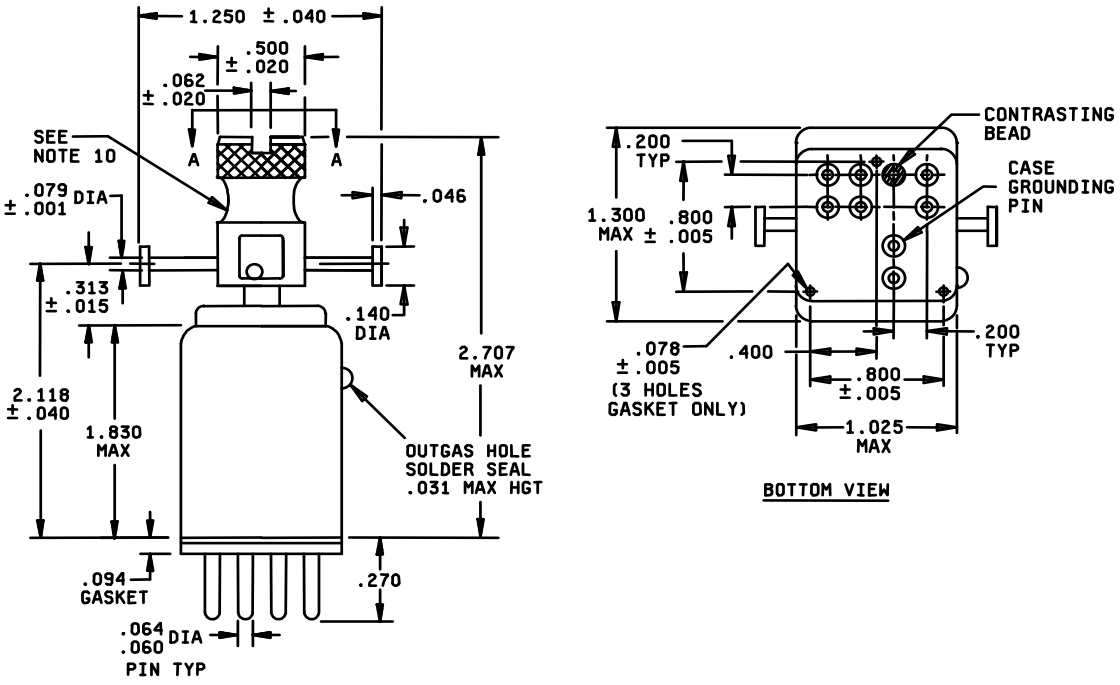
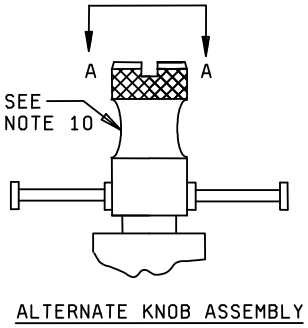



FIGURE 1. Dimensions and configuration.

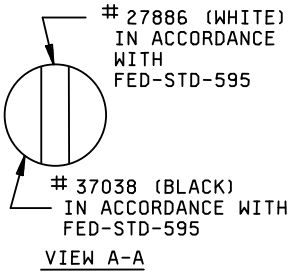


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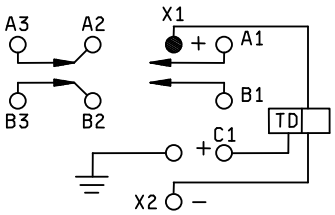
| Inches | mm |
|---|---|
| .001 | 0.03 |
| .005 | 0.13 |
| .015 | 0.38 |
| .031 | 0.79 |
| .040 | 1.02 |
| .046 | 1.17 |
| .060 | 1.52 |
| .064 | 1.63 |
| .078 | 1.98 |
| .079 | 2.01 |
| .094 | 2.39 |
| .140 | 3.56 |
| .200 | 5.08 |
| .270 | 6.86 |
| .313 | 7.95 |
| .400 | 10.16 |
| .800 | 20.32 |
| 1.025 | 26.04 |
| 1.250 | 31.75 |
| 2.118 | 53.80 |
| 2.707 | 68.76 |




ALTERNATE KNOB ASSEMBLY




VIEW A-A




CIRCUIT DIAGRAM

TERMINAL VIEW

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. Terminal numbers shall not appear on the relay header. There shall be a legible circuit diagramon the relay which identifies each terminal location specified.
5. For mounting track, see MIL-PRF-12883/49. For bracket and socket assembly, see MIL-PRF-12883/51.
6. Gasket material: Methylvinyl silicone; shore hardness A40.
7. Terminal plating: Shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold plated contacts. Terminals shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microninches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
8. Enclosure finish color: FED-STD-595, green 34449.
9. Coil suppression: Back EMF 42 V dc maximum.
10. Knob and post shall be nickel plated (100 microinches thick minimum), 303 stainless steel.

FIGURE 1. Dimensions and configuration - Continued.



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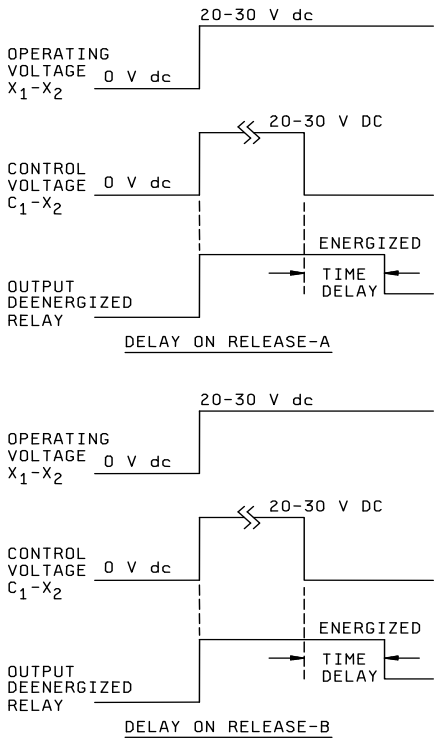



FIGURE 1. Dimensions and configuration - Continued.

TABLE I. Dash numbers and time delay characteristics.

| Dash number 1/ 2/ | Time delay (seconds) ±10% 3/ |
|---|---|
| -1000 | 0.1 |
| -1001 | 1.0 |
| -5001 | 5.0 |
| -1602 | 16.0 |
| -6002 | 60.0 |
| -1803 | 180.0 |
| -3003 | 300.0 |

1/ Additional time delay relays within the 0.1- to 300-second delay range are available. To establish Part or Identifying numbers (PIN's) not listed in table I (see "PIN" herein).
2/ PIN's having a suffix letter A following the dash number impose more stringent EMI requirements than PIN's with a suffix letter W. For Government logistical support the suffixed PIN's will be supplied (see "PIN" herein).
3/ Ten percent of the nominal value under all combinations of input voltage and environmental extremes.



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REQUIREMENTS:

Contact data:

Configuration: 2PDT.

Life/load ratings: See table II.

TABLE II. Life load ratings (relay case grounded).

| Type of load | Life (cycles) | Amperes | |
|---|---|---|---|
| | | 28 V dc | 115/200 1 and 3 phase 400 Hz |
| Resistive | 100,000 | 10 | 10 |
| Inductive | 20,000 | 8 | 8 |
| Motor | 100,000 | 4 | 4 |
| Lamp | 100,000 | 2 | 2 |
| Mechanical life (reduced current) | 400,000 | 2.5 | 2.5 |

Contact voltage drop:

Initial: 0.150 volt.

After life tests: 0.175 volt.

Intermediate current: Applicable in accordance with MIL-PRF-83536.

Contact bounce: 1 millisecond (maximum).

Overload:

DC: 40 amperes.

AC: 60 amperes.

Rupture: 1/

DC: 50 amperes.

AC: 80 amperes.

Input data:

Duty rating: Continuous.

Maximum voltage (over temperature range): 30 V dc.

1/ Not applicable if time current relay characteristics at 25°C are required, or if the MIL-PRF-83536/9,/10, /11 being used is qualified to MIL-PRF-83536.



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Rated voltage (over temperature range): 28 V dc.

Minimum voltage (over temperature range): 20 V dc.

Minimum voltage (high-temperature test): 20 V dc.

Minimum voltage (continuous current tests): 20 V dc.

Maximum coil current at 25°C: 0.150 ampere.

Operate time: 10 milliseconds (maximum).

Recycle time: 50 milliseconds maximum (over the total temperature range and with both minimum and maximum input voltages). Relay will recycle if control voltage is removed from and reapplied to the control terminal C1 during the timing cycles.

Electrical data:

Insulation resistance at 500 V dc: 2/

Initial: 100 megohms (minimum).

After life or environmental tests: 100 megohms (minimum).

Dielectric strength (sea level): 2/

| | Initial | After life tests |
|---|---|---|
| Coil to case. 3/ | 1,050 V rms | 1,050 V rms |
| All other points. 4/ | 1,250 V rms | 1,250 V rms |

Dielectric strength (altitude)
(80,000 feet): 2/ 5/

| | Initial | After life tests |
|---|---|---|
| Coil to case. 3/ | 350 V rms | 350 V rms |
| All other points. 4/ | 350 V rms | 350 V rms |

Transients (at terminals X1, X2, and C1):

Transient voltage limits (input): In accordance with MIL-STD-461 (methods CS101, CS114, RS101, and RS103) for PIN's with a suffix letter W. For PIN's with a suffix letter A, the same limits apply except that the field intensity limits for RS103 are as specified in table III. Susceptibility testing shall not cause the relay to change state nor affect the timing interval beyond allowable tolerance limits.

2/ Terminals X1, X2 and C1 are connected together during this test.
3/ Leakage current is limited to 50 microamperes.
4/ Leakage current is limited to 25 microamperes.
5/ Relay is not installed in its bracket and socket assembly (MIL-PRF-12883/51-004).



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TABLE III. RS103 levels for suffix letter A PIN's.

| Frequency | RS103 level (volt/meter) |
|---|---|
| 14 kHz to 200 MHz | 20 |
| 200 MHz to 450 MHz | 65 |
| 450 MHz to 1 GHz | 20 |
| 1GHz to 12 GHz | 100 |

Transient voltage limits (output): In accordance with MIL-STD-461 (methods CE106 and RE103), voltage spikes generated by time delay relays shall not exceed 42 volts during the following time periods starting at the instant of opening of the relay coil circuit:

PIN's with a suffix letter W: 50 microseconds.

PIN's with a suffix letter A: 500 microseconds.

Monitoring of spike voltages shall be by a device (such as a memory type oscilloscope) that can detect, measure, display, and record 0.25 to 1000 volts over the applicable time durations specified herein. Switches used to control the time delay relays during voltage-spike measurements shall not be EMI suppressed.

Time current relay characteristics at 25°C: 6/ Relays shall sustain five applications (make and carry only) of power concurrently on adjacent poles at each of five different current levels for the time durations shown in table IV. Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase. Cooling time between successive applications shall be 30 minutes. The test shall be performed on both normally open and normally closed contacts of each relay. There shall be no failures or evidence of welding or sticking, and relays shall pass contact voltage drop at conclusion.

TABLE IV. Time-current relay characteristics at 25°C.

| | | |
|---|---|---|
| 1 | 15 A | 1 hour |
| 2 | 59 A | 5.0 seconds |
| 3 | 100 A | 1.2 seconds |
| 4 | 250 A | 0.2 second |
| 5 | 350 A | 0.1 second |

CAUTION: Compare with time-current characteristics of associated circuit protective device.

Environmental data (relay installed in bracket and socket assembly M12883/50 and secured in mounting track M12883/49.

Temperature range (operating): -70°C to +125°C.

Maximum altitude rating: 80,000 feet.

6/ Not applicable if the MIL-PRF-83536/9, /10, /11 relay being used is qualified to the time current relay characteristics at 25°C requirements of MIL-PRF-83536.



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Shock G level: 50 g's.

Duration: 10 milliseconds.

Maximum duration contact opening: 10 microseconds.

Maximum duration contact closing: 1 microsecond.

Vibration (sinusoidal):

G level: 30 g's.

Frequency range: 10 Hz to 3,000 Hz.

Vibration (random): Applicable in accordance with MIL-STD-202, method 214, test condition 1B.

Power spectral density: 0.4 $g^2$/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Duration: 15 minutes each plane.

Acceleration: 15 g's.

Seal: Applicable to nonhermetically-sealed relays and hermetically-sealed (potted) relays.

Physical data:

Dimensions and configuration: See figure 1.

Weight: 0.34 pound (.153 kilogram) maximum.

Knob strength: The rod to knob junction shall be capable of resisting an axial pull of 35 pounds without degradation.

Marking: See MIL-PRF-83726. In addition, relays shall be marked with the ESDS identifier as specified in MIL-STD-1285.

ESDS protection program: The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan that must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the EDS/ESD Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

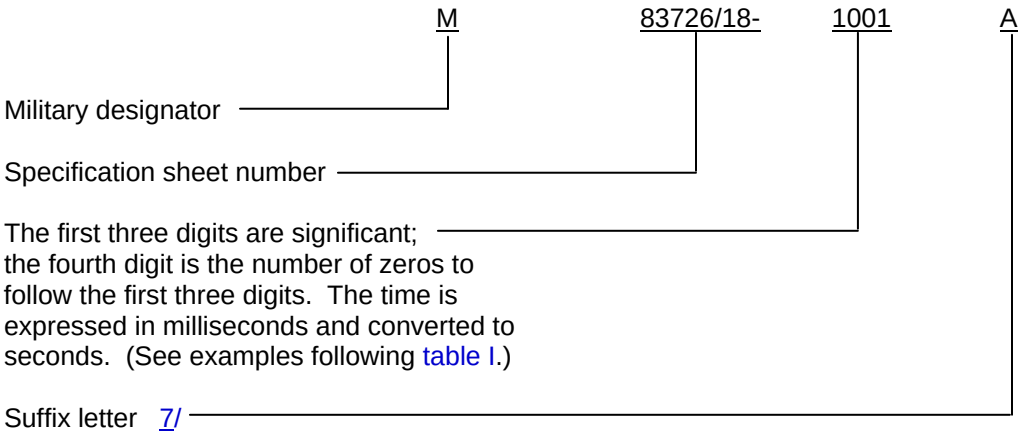
ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.



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ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

PIN: Consists of the prefix M83726/18-, a four-digit dash number (expressed in milliseconds), and a suffix letter (indicating the extent of EMI requirements) as follows:




Examples:

M83726/18-1000W-100 millisecond time delay, with less stringent EMI requirements.

M83726/18-1001A-1 second time delay, with more stringent EMI requirements.

Quality assurance provisions: Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes, and shall impose additional testing requirements as necessary.

The Qualified Products List (QPL) associated with this inactive for new design specification will be maintained until acquisition of the product is no longer required, whereupon the specification and the QPL will be canceled.

7/ Any relays numbered prior to the date of this specification with a suffix indication shall be considered interchangeable (store and issue) with the W for relays with less stringent EMI requirements (Y,X) or A for relays with more stringent EMI requirements (B,C).



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NOTE: As of 15 June 1999, MIL-PRF-83726 no longer specifies Quality Levels, but existing order configurations may still include them. Relays with a "W" quality level indicator at the end are interchangeable with relays without a quality level letter.

Referenced documents. In addition to MIL-PRF-83726, this document references the following:

| | | |
|---|---|---|
| FED-STD-595/27886 | FED-STD-595/34449 | FED-STD-595/37038 |
| MIL-STD-202-214 | MIL-STD-461 | MIL-STD-883 |
| MIL-STD-1285 | MIL-PRF-12883/49 | MIL-PRF-12883/51 |
| MIL-STD-1686 | MIL-PRF-83536 | MIL-PRF-83536/9 |
| MIL-PRF-83536/10 | MIL-PRF-83536/11 | ASTM B488 |

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Air Force - 85
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Preparing activity:
DLA - CC

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